Problem

The following results were obtained from measurements taken between the two terminals of a resistive network.

Terminal Voltage	72 V	0 V
Terminal Current	0 A	9 A

Find the Thevenin equivalent of the network.

Step-by-step solution

Step 1 of 2

The Thevenin's voltage is equal to the open circuit voltage, that is, the value of terminal voltage when the terminal current is 0.

The terminal current $i = 0\text{ A}$ corresponds with $V_{\text{Th}} = 72\text{ V}$.

The Norton's current is equal to the short circuit current, that is, the value of terminal current when the terminal voltage is 0.

The terminal voltage $v = 0\text{ V}$ corresponds with $I_{\text{N}} = 9\text{ A}$.

The Thevenin's resistance (R_{Th}) is represented as the ratio of Thevenin's voltage to the Norton's current.

$$R_{\text{Th}} = \frac{V_{\text{Th}}}{I_{\text{SC}}}$$

Substitute 9 A for I_{SC} and 72 V for V_{Th}.

$$\begin{aligned} R_{\text{Th}} &= \frac{72}{9} \\ &= 8\ \Omega \end{aligned}$$

Hence, the Thevenin's resistance (R_{Th}) is $8\ \Omega$.

Step 2 of 2

The Thevenin’s equivalent network is represented as the Thevenin’s voltage (V_{Th}) in series with the Thevenin’s resistance (R_{Th}) with the load terminal open is shown in Figure 1.

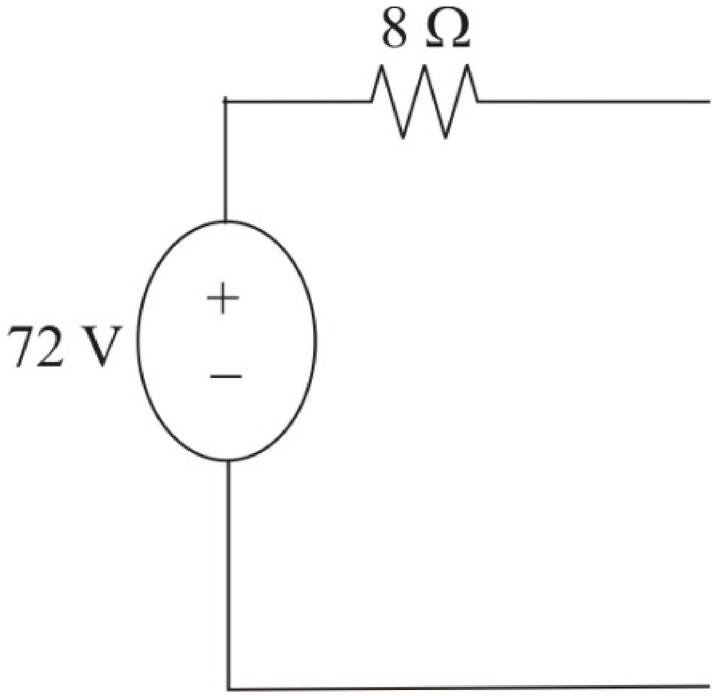

Figure 1

Hence, the required Thevenin’s network has been shown in Figure 1.